US Patent & Trademark Office
Patent Public Search | Text View

United States Patent 12404920
Kind Code B2
Date of Patent September 02, 2025
Inventor(s) Ogrizovic; Goran

Drive device for a car, in particular for a passenger car

Abstract

A drive device for a car includes a first output shaft via which a first vehicle wheel of the car is drivable, a second output shaft arranged coaxially to the first output shaft via which a second vehicle wheel of the car is drivable, a differential transmission via which the output shafts are drivable, and three axial flux machines arranged coaxially to one another and coaxially to the output shafts. The three axial flux machines are a first axial flux machine via which the first output shaft is drivable by bypassing the differential transmission, a second axial flux machine via which the second output shaft is drivable by bypassing the differential transmission, and a third axial flux machine between the first axial flux machine and the second axial flux machine in the axial direction of the axial flux machines via which the output shafts are drivable via the differential transmission.

Inventors: **Ogrizovic; Goran (Nuertingen, DE)**

Applicant: **Mercedes-Benz Group AG** (Stuttgart, DE)

Family ID: **84901152**

Assignee: **Mercedes-Benz Group AG (Stuttgart, DE)**

Appl. No.: **18/722191**

Filed (or PCT Filed): **December 19, 2022**

PCT No.: **PCT/EP2022/086745**

PCT Pub. No.: **WO2023/126237**

PCT Pub. Date: July 06, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250052306 A1	Feb. 13, 2025

Foreign Application Priority Data

DE	10 2022 000 018.0	Jan. 03, 2022

Publication Classification

Int. Cl.: **F16H37/08** (20060101); **B60K1/02** (20060101); **B60K17/16** (20060101); **B60K17/26** (20060101)

U.S. Cl.:

CPC **F16H37/0806** (20130101); **B60K1/02** (20130101); **B60K17/165** (20130101); **B60K17/26** (20130101);

Field of Classification Search

CPC: F16H (37/0806); B60K (1/02); B60K (2001/001); B60K (17/165); B62D (11/14)

USPC: 475/18

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8813879	12/2013	Walter	180/69.6	H02P 25/22
8992367	12/2014	Kalmbach	475/150	H02K 7/006
9067484	12/2014	Zhao	N/A	B60K 1/00
9255633	12/2015	Märkl	N/A	B60K 17/165
10703200	12/2019	Stoermer	N/A	N/A
11005337	12/2020	Hung	N/A	B60K 17/165
2015/0107914	12/2014	Zhao	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
113085518	12/2020	CN	N/A
10 2015 000 466	12/2015	DE	N/A
10 2019 115 918	12/2019	DE	N/A
102020209431	12/2021	DE	N/A
WO 2018/150010	12/2017	WO	N/A

OTHER PUBLICATIONS

International Search Report (PCT/ISA/210) issued in PCT Application No. PCT/EP2022/086745 dated May 10, 2023 (2 pages). cited by applicant

---

Primary Examiner: Knight; Derek D

Attorney, Agent or Firm: Crowell & Moring LLP

Background/Summary

BACKGROUND AND SUMMARY OF THE INVENTION

(1) The invention relates to a drive device for a car, in particular for a passenger car.

(2) A drive device for a motor vehicle is known from DE 10 2015 000 466 B4. Moreover, WO 2018/150010 A1 discloses an axis drive unit for a motor vehicle that can be driven electrically.

(3) The object of the present invention is to create a drive deice for a car, such that a particularly advantageous drive of the car can be achieved.

(4) The invention relates to a drive device, also referred to as an axis drive, for a car, in particular for a passenger car. This means that, in its completely produced state, the car has the drive device and can be driven by means of the drive device, in particular purely electrically. In particular, the car has at least one vehicle axis, also simply referred to as an axis, which has at least or exactly two vehicle wheels, also referred to simply as wheels. As is explained below in more detail, the vehicle wheels can be driven by means of the drive device, in particular purely electrically, whereby the car can be driven, in particular purely electrically. In particular, the vehicle wheels are arranged on sides of the car, also referred to as a vehicle, that lie opposite each other in the transverse direction of the vehicle.

(5) The drive device has a first output shaft, by which or via which a first of the vehicle wheels of the vehicle axis can be driven. To do so, for example, the first vehicle wheel is coupled or can be coupled to the first output shaft in a manner that transfers torsional moments. The drive device has a second output shaft arranged coaxially to the first output shaft, by which or via which the second vehicle wheel of the vehicle axis of the car can be driven. To do so, for example, the second vehicle wheel is coupled or can be coupled to the second output shaft in a manner that transfers torsional moments. Moreover, the drive device comprises a differential transmission, via which the output shafts can be driven. In other words, the output shafts can be driven by the differential transmission. The differential transmission is formed, for example, as a cone differential, which is also referred to as a cone wheel differential. Of course it is conceivable that the differential transmission is formed as a different kind of differential transmission. As is already well known from the general prior art, the differential transmission is formed, in particular, to allow different rotational speeds of the output shafts and thus of the vehicle wheels when cornering the car, for example, in particular in such a way that the curve outer vehicle wheel rotates or can rotate with a higher rotational speed than the curve inner vehicle wheel, in particular while the vehicle wheels can be driven or are driven by means of the drive device. Moreover, the drive device has three axial flux machines arranged coaxially to one another and coaxially to the output shafts, namely a first axial flux machine, a second axial flux machine and a third axial flux machine. The respective axial flux machine has a respective stator and, in each case, at least or exactly two rotors, which can be driven by means of the respective stator and thus can be rotated around a respective engine axis of rotation in relation to the respective stator. Here, the respective rotor of the respective axial flux machine is arranged between the respective rotors of the respective axial flux machine in the axial direction of the respective axial flux machine, wherein the respective axial direction of the respective axial flux machine coincides with the respective engine axis of rotation. In particular, the axial flux machines are arranged following on from one another and thus one behind the other in the axial direction of the respective axial flux machine, in particular completely. The rotors of the first axial flux machine are connected to one another in a rotationally fixed manner and thus form a first complete rotor. The rotors of the second axial flux machine are connected to one another in a rotationally fixed manner and thus form a second complete rotor. The rotors of the third axial flux

machine are connected to one another in a rotationally fixed manner and thus form a third complete rotor.

(6) By bypassing the differential transmission, the first output shaft can be driven by means of the first axial flux machine, in particular by means of the complete rotor of the first axial flux machine. This means that, based on a first torsional moment flow, via which a respective first torsional moment can be transferred from the complete rotor of the first axial flux machine to the first output shaft, in order to thus drive the first output shaft, the differential transmission is not arranged in the first torsional moment flow or at least not in the first torsional moment flow between the complete rotor of the first axial flow machine and the first output shaft, such that the respective first torsional moment does not flow via the differential transmission on its way from the complete rotor of the first axial flux machine to the or onto the first output shaft along the torsional moment flow, and thus bypasses the differential transmission. By bypassing the differential transmission, the second output shaft can be driven by means of the second axial flux machine, in particular by means of the complete rotor of the second axial flux machine. This means that, based on a second torsional moment flow, via which a respective second torsional moment can be transferred from the complete rotor of the second axial flux machine to the or onto the second output shaft, the differential transmission is not arranged in the second torsional moment flow or, in any case, not in the second torsional moment flow between the complete rotor of the second axial flux machine and the second output shaft. Thus, the respective second drive torsional moment on its way from the complete rotor of the second axial flux machine to the or onto the second output shaft along the torsional moment flow, not flowing via the differential transmission, and thus bypasses the differential transmission. Expressed in other words again, the first or second axial flux machine can provide the respective first or second torsional moment via its complete rotor, in order to thus drive the first or second output shaft. The respective first or second torsional moment is here transferred via the first or second torsional moment flow from the complete rotor of the first or second axial flux machine onto the or to the first or second output shaft, wherein when it flows along the first or second torsional moment flow from the complete rotor of the first or second axial flux machine to the or onto the first or second output shaft and thus has reached the first or second output shaft, the respective first or second torsional moment is not flowed via the differential transmission from the complete rotor of the first or second axial flux machine onto the or to the first or second output shaft.

(7) The output shafts can be driven via the differential transmission by means of the third axial flux machine, in particular by means of the complete rotor of the third axial flux machine. This means that the third axial flux machine can provide a respective third torsional moment via its complete rotor for driving the respective output shaft. Based on a third torsional moment flow via which the respective third torsional moment can be transferred from the complete rotor of the third axial flux machine to the respective output shaft, the differential transmission is arranged in the third torsional moment flow and here upstream of the respective output shaft and downstream of the complete rotor of the third axial flux machine, such that the respective third torsional moment on its way from the complete rotor of the third axial flux machine to the or onto the respective output shaft via the differential transmission flows from the complete rotor of the third axial flux machine onto the or to the respective output shaft. To do so, it is provided, for example, that the complete rotor of the third axial flux machine is connected to an input element of the differential transmission, in particular in a rotationally fixed manner, wherein the input element of the differential transmission, for example, which is also simply referred to as a differential, is a differential cage.

(8) In an advantageous design of the invention, a first freewheel is provided, via which the first output shaft of the first axial flux machine can be driven by bypassing the differential transmission. A second freewheel is also provided, via which the second output shaft of the second axial flux machine can be driven by bypassing the differential transmission.

(9) In an advantageous design of the invention, a first frictionally engaged or positive-locking switching element is provided, by means of which the first output shaft of the first axial flux machine can be driven by bypassing the differential transmission. A second frictionally engaged or positive-locking switching element is also provided, by means of which the second output shaft of the second axial flux machine can be driven by bypassing the differential transmission.
(10) In particular, the invention is based on the following knowledge and considerations: the drive device according to the invention is based on a so-called dual motor drive unit, since a respective electric engine is presently provided in the form of the first electric engine or the second electric engine per output shaft and thus per output shaft side. In doing so, a particularly advantageous driving dynamic can be achieved, since a torsional moment distribution, also referred to as torque vectoring, in particular, and other driving dynamic interventions can be achieved. Based on this dual motor drive unit, the dual motor drive unit is augmented by a third electric engine, wherein all electric engines are formed as axial flux machines, whereby in particular a particularly advantageous support operation, also referred to as boost operation or boost mode, can be achieved, wherein by forming the electric engines as axial flux machines, the necessary axial construction space can be kept low. Vehicles that are designed with a dual motor drive unit are generally vehicles with a high performance. This high performance is provided by high-performance electric engines, which, however, cannot usually be optimally designed for a normal driving cycle, in which high performance is not required. Smaller electric engines are generally more efficient in conventional everyday driving cycles. Here, the goal is to depict the functionality of a dual motor drive unit in the vehicle, however as far as possible without efficiency disadvantages. In particular with an axial flux machine, the radial scaling for performance torsional moment increase has a negative impact on the permissible maximum rotational speed. Since the radial construction space for electric engines is generally limited and thus also the performance or the torsional moment per electric engine, the third electric engine allows an advantageous performance scaling. Depending on the radial dimensioning of the respective complete rotor of the respective axial flux machine, the rotational speed limits can emerge. Smaller rotor dimensions allow higher rotational speeds. As a result of corresponding motor dimensioning, the two electric engines bound to the output shafts can rotate more quickly than the third axial flux machine used as a boost electric engine, and thus a higher maximum speed of the car can be achieved. The third axial flux machine also referred to as a power electric engine could thus be decoupled above a certain speed. In other words, based on a dual motor drive unit, the third axial flux machine is used as a boost or power electric motor or electric engine, which is bound to the differential transmission, from which the output shafts lead out, in particular as differential shafts, on which the first axial flux machine and the second axial flux machine can have an impact. In particular, it is conceivable that the first or second axial flux machine can be switched via a respective transfer element, such as a freewheel or via an actuated or actuatable switching element, such has a frictional coupling, for example, or a claw coupling or can act on the respective output shaft, in particular by bypassing the differential transmission.
(11) The invention enables a particularly efficient operation with a particularly high performance density, since the three axial flux machines can be arranged coaxially. In particular, an advantageous scalability can thus be depicted. Rotors of axial flux machines with smaller diameters can be used, which enables a higher maximum rotational speed, with which single-start transmissions with high highest speed can be depicted better.
(12) Further advantages, features and details of the invention emerge from the below description of preferred exemplary embodiments and by means of the drawings. The features and feature combinations mentioned above in the description and the features and feature combinations mentioned below in the description of the figures and/or shown only in the figures can be used not only in the respectively specified combination, but also in other combinations or on their own, without leaving the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** a schematic depiction of a first embodiment of a drive device for a car, in particular for a passenger car; and
(2) FIG. **2** a schematic depiction of a second embodiment of the drive device.
DETAILED DESCRIPTION OF THE DRAWINGS
(3) In a schematic depiction, FIG. **1** shows a first embodiment of a drive device **10** for a car, in particular for a passenger car. The car is also referred to as a motor vehicle or vehicle and has, for example, at least or exactly two vehicle axes arranged one after the other in the longitudinal direction of the vehicle and also simply referred to as axes. The respective vehicle axis has at least or exactly two vehicle wheels, wherein the respective vehicle wheels of the respective vehicle axis are arranged on sides of the motor vehicle lying opposite one another in the transverse direction of the vehicle. As is explained in yet more detail below, the vehicle wheels of one of the vehicle axes can be driven purely electrically, in particular, by means of the drive device **10**. When we speak below of vehicle wheels, unless otherwise specified, this is to be understood to mean the vehicle wheels of the one vehicle axis that can be driven by means of the drive device **10**. The drive device **10** has a first output shaft **12**, via which a first of the vehicle wheels, labelled with **14** in FIG. **1**, can be driven. Furthermore, the drive device **10** has a second output shaft **16**, via which the second vehicle wheel, labelled with **18** in FIG. **1**, can be driven. Furthermore, the drive device **10** has a differential transmission **20**, which is formed as a cone wheel differential in the first embodiment and, here, has a differential cage **22** as the input element. The output shafts **12** and **16** are arranged coaxially to each other and can be rotated around a common output shaft axis of rotation in relation to a housing **24** of the drive device **10**. Here, the differential transmission **20**, which is also simply referred to as the differential, is arranged coaxially to the output shafts **12** and **16** and thus can be rotated around the output shaft axis of rotation in relation to the housing **24**. Here, the differential cage **22**, in particular, can be rotated around the output shaft axis of rotation in relation to the housing **24**. Compensation wheels **26** are arranged on the differential cage **22**, the compensation wheels being able to be rotated around a common compensation wheel axis of rotation in relation to the differential cage **22** and also in relation to one another, for example. The compensation wheel axis of rotation here runs perpendicularly to the output shaft axis of rotation. In the first embodiment, the respective compensation wheel **26** is formed as a tooth wheel and, here, in particular as a bevel wheel. A first output wheel **28** of the differential is connected to the output shaft **12** in a rotationally fixed manner, and a second output wheel **30** of the differential transmission **20** is connected to the output shaft **16** in a rotationally fixed manner. The output wheels **28** and **30** are tooth wheels, in particular bevel wheels, wherein the respective output wheel **28**, **30** meshes with the compensation wheels **26**. Thus, the output wheels **28** and **30** can be rotated along with the output shafts **12** and **16** around the output shaft axis of rotation in relation to the housing **24**.
(4) Furthermore, the drive device **10** has three axial flux machines **32***a-c* arranged coaxially to one another and coaxially to the output shafts **12** and **16** and coaxially to the differential transmission **20**. Here, the axial flux machine **32***a* is referred to as the first axial flux machine, the axial flux machine **32***b* as the second axial flux machine and the axial flux machine **32***c* as the third axial flux machine. It can be seen that the third axial flux machine **32***c* is arranged between the axial flux machines **32***a* and **32***b* in the axial direction of the axial flux machines **32***a-c*. The respective axial flux machine **32***a-c* has a respective stator **34***a-c*, which is connected to the housing **24** in a rotationally fixed manner. Furthermore, the respective axial flux machine **32***a-c* respectively has two rotors **36**, **38** or **40**. Here, the stator **34***a* is arranged between the rotors **36** of the axial flux machine **32***a* in the axial direction of the axial flux machine **32***a*. Correspondingly, the rotor stator

34*b* of the axial flux machine **32***b* is arranged between the rotors **38** of the axial flux machine **32***b* in the axial direction of the axial flux machine **32***b*. Accordingly, the stator **34***c* of the axial flux machine **32***c* is arranged between the rotors **40** of the axial flux machine **32***c* in the axial direction of the axial flux machine **32***c*.

(5) The rotors **40** are connected to the differential cage **22** in a rotationally fixed manner, such that the differential transmission **20** can be driven by the rotors **40**. As a result, the output shafts **12** and **16** can be driven by the rotors **40** and thus by the axial flux machine **32***c* via the differential transmission **20**. By driving the output shafts **12** and **16**, the vehicle wheels **14** and **18** are driven. The output shaft **12** can be driven by the rotors **36** and thus by the axial flux machine **32***a* by bypassing the differential transmission **20**, and the output shaft **16** can be driven by the rotors **38** and thus by the axial flux machine **32***b* by bypassing the differential transmission **20**. Here, the drive device **10** has a first transfer element **42** allocated to the output shaft **12** and the rotors **36** and a second transfer element **44** allocated to the output shaft **16** and rotors **38**. The output shaft **12** can be driven via the transfer element **42** by the rotors **36** by bypassing the differential, and the output shaft **16** can be driven via the transfer element **44** by the rotors **38** by bypassing the differential transmission **20**. For example, the transfer element **42**, **44** is formed as a freewheel. Furthermore, it is conceivable that the respective transfer element **42**, **44** is formed as a switching element, in particular that can be actuated, which can be as a frictionally engaged switching element, such as a frictional coupling, for example, in particular a lamella coupling, or even a positive-locking switching element, such as a claw coupling, for example. A respective bearing of the respective output shaft **12**, **16** is labelled in FIG. **1** with **46** or **48**.

(6) Moreover, it can be seen from FIG. **1** that a respective planetary gearset **50**, **52** is allocated to the respective output shaft **12**, **16**. The respective planetary gearset **50**, **52** has a respective sun gear **54**, a respective planetary carrier **56** and a respective gear ring **58**. For example, the gear ring **58** is connected to the housing **24** in a rotationally fixed manner. Furthermore, the respective planetary gearset **50**, **52** has respective planetary wheels **60**, which are mounted rotatably on the respective planetary carrier **56** of the respective planetary gearset **50**, **52**. The respective sun gear **54** is connected to the respective output shaft **12**, **16** in a rotationally fixed manner. Moreover, the respective vehicle wheel **14**, **18** is connected to the respective planetary carrier **56** in a manner transferring torsional moment. The axial flow machine **32***c* arranged between the axial flow machines **32***a* and **32***b* is a central electric engine, which is also referred to as a main motor or efficiency motor and is bound to the differential. For example, the axial flux machines **32***a* and **32***b* are so-called boost motors. The output shafts **12** and **16** formed as differential shafts lead out of the differential transmission **20** to transmissions which are presently formed as planetary gearsets **50** and **52**. It can be seen that the output shafts **12** and **16** penetrate the rotors **36** and **38** or respective rotor shafts of the rotors **36** and **38** and respective hub regions of the axial flux machines **32***a* and **32***b*. The respective boost motor **32***a* or **32***b* provided per side can be coupled or is coupled to the respective differential shaft, in each case via a transfer element **42**, **44** formed as a freewheel or a switching element. A respective freewheel can be integrated in the respective hub region of the respective boost motor **32***a, b*. The freewheels are, for example, either in the hub bore, as is depicted, or engage outwardly on the hub on a shaft section. These freewheels have, for example, a centrifugal-activated clamp installation. A hydrodynamic clamp removal is also conceivable. With zero rotational speed or low absolute rotational speed of the respective boost motor, which is also referred to as the boost electric engine, the freewheel clamps do not abut on the shaft or the hub bore. The reduces the friction and the attrition of the clamping bodies. Above a certain speed, the clamping bodies are in contact and can develop their locking effect. The locking effect occurs as soon as the rotational speed of the respective boost motor reaches or exceeds the rotational speed of the differential shafts. Only when both sides with the respective freewheel are in the locked state can a torsional moment of the boost motors distributed evenly on both sides be transferred. The locking of the freewheels can also selectively take place if a torsional moment engagement is

wanted on only one side. Such a freewheel would be conceivable as a transfer element **42** or **44**. Alternatively to freewheels, in each case a switchable, i.e., actuatable, switching element, for example in the form of a frictional coupling or a claw coupling, can be used as the transfer elements **42**, **44**. The boost motors **32***a*, *b* only generate the respective torsional moment in one direction and, for example, when driving forward, if only freewheels are used as transfer elements **42**, **44**. If switching elements are used as transfer elements **42**, **44**, the respective boost motor **32***a*, *b* can also be operated in reverse. When using freewheels as the transfer elements **42**, **44**, the following can be provided, in particular: the two axial flux machines **32***a* and **32***b* formed as boost motors, for example, can generate propulsion moments independently of one another, in particular when driving forwards. With switched-on boost motors **32***a*, *b*, the torsional moment distribution between the two wheels **14**, **18** is regulated via these two boost motors **32***a*, *b*. When using switching elements the following can apply: the two boost motors **32***a*, *b* can generate propulsion moments or braking moments independently of each other, in particular when driving forwards or backwards. With switched-on boost motors **32***a*, *b*, the torsional moment distribution between the two wheels **14**, **18** is regulated via these two boost motors **32***a*, *b*. In general, it is as follows that, depending on a radial dimensioning of the rotor and the rotor design of an axial flux machine rotor, rotational speed limits emerge, smaller rotor dimensions enable higher rotational speeds. The boost motors **32***a*, *b* are thus advantageously decoupled above a certain speed. The highest speed corresponding to the entire installed electric engine performance of the three axial flux machine **32***a*-*c* on this axis would, in the event of a non-decoupling, not be able to be achieved, and the front axis of the vehicle would have to optionally provide the necessary power for the highest speed. The object of the present invention is to develop an efficient electrical drive system for moderate driving requirements, which is not over-dimensioned for these requirements. An efficiency-optimized electric engines **32***c* can be used, which generally also has a lower performance and a lower maximum torsional moment. The boost motors **32***a*, *b* are only still used in the comparatively rare cases of a high performance requirement or a moment engagement on the respective wheel **14**, **18**. The losses, which would emerge as a result of the electric drag moment of a boost motor **32***a*, *b* formed, in each case, as a permanently energized electric engine, and the bearing losses, which would emerge in the bearings of the respective rotor, do not apply. In principle, by using axial flux machines, very high performance densities and torsional moment densities can be achieved. When using centrifugal-activated freewheels as transfer elements **42**, **44**, the output shafts should already be rotated or the vehicle should have a minimum speed or be in the acceleration process, since the centrifugal-activated clamps only make contact above a certain rotational speed and can only then lock. With an acceleration from a standstill or from a low speed, no noticeable switching of the boost motors **32***a*, *b* emerge. When using switching elements as transfer elements **42**, **44** and conventional freewheels with constantly abutting clamping bodies, these limitations do not exist.

(7) In a schematic depiction, FIG. **2** shows a second embodiment of the drive device **10**. Here, the differential transmission **20** has differential shafts **62** and **64** provided, in particular, in addition to the output shafts **12** and **16**, the differential shafts being arranged coaxially to one another and coaxially to the output shafts **12** and **16**. Here, the output wheel **28** is connected to the differential shaft **62** in a rotationally fixed manner and the output wheel **30** is connected to the differential shaft **64** in a rotationally fixed manner. The rotors **36** and, via these, the output shaft **12** can be driven via the transfer element **42** by the differential shaft **62** and thus by the axial flux machine **32***c*, such that the axial flux machine **32***c* can drive the rotors **36** via the differential transmission **20** and the output shaft **12** via these. Via the transfer element **44**, the differential shaft **64** can drive the rotors **38** and, via these, the output shaft **16**, such that the axial flux machine **32***c* can drive the rotors **38** via the differential transmission **20** and, via these, the output shaft **16**. Bearings for the differential shafts **62** and **64** are labelled with **66** and **68**. The respective transfer element **42**, **44** can be, for example, a freewheel or an in particular actuatable switching element, such as a frictionally engaged switching element or a positive-locking switching element, for example. Here, for

example, the axial flux machines **32***a* and **32***b* are formed as efficiency motors or efficiency engines, wherein the axial flux machine **32***c*, in particular, is formed as a boost motor, i.e., as a boost engine. In particular, the axial flux machines **32***a* and **32***b* are electric engines with relatively low power and torsional moment, wherein the axial flux machines **32***a* and **32***b* are permanently bound to the vehicle wheels **14** and **18**. The boost motor **32***c* is switched, for example when driving forwards, via the respective transfer element **42**, **44** via the respective differential shaft **62**, **64** onto the output. The respective transfer element **42**, **44** is, for example, in each case a centrifugal-activated freewheel, in which, as a result of the clamp removal, no abutment of the clamping bodies is present when the boost motor **32***c* is inactive, and thus no additional drag losses and attrition are generated on the freewheel. Optionally, the axial flux machine **32***c* can be formed as a performance engine, in particular with maximum performance requirements. Wheel-individual drive torsional moment and braking torsional moment of the two efficiency engines **32***a, b* immediately make contact. The performance engine **32***c* has low switching times, since no active switching element is actuated and no synchronization of a switching element is required when the transfer elements **42** and **44** are advantageously formed as freewheels. In the event of a one-sided slip, a high torsional moment can also advantageously be portrayed on the opposite, non-slipping wheel, in particular a maximum torsional moment of the respective efficiency engine **32***a, b* in addition to a half available torsional moment of the performance engine **32***c*, wherein the respective efficiency engine **32***a, b* of a slipping side is additionally supported or braked as needed. Torsional moment requirements, performance requirements and highest speed requirements can thus advantageously be depicted with one gear with the embodiment of the invention from FIG. **2**. The ability of the boost motor to decouple to the benefit of the highest speed is thus advantageously possible with the embodiment of the invention from FIG. **2** as well as with transfer elements **42**, **44** designed as the freewheel, if the maximum rotational speed of the efficiency engine is higher than the maximum rotational speed of the boost motor. In this operating situation, the highest speed is only driven with the two efficiency engines, since the boost motor remains decoupled. In particular on a rear axis, particularly high performance can thus be achieved.
(8) Furthermore, it is conceivable that the axial flux machine **32***c* is formed as an efficiency engine with relatively low performance and torsional moment and, in particular as in the first embodiment of the invention in FIG. **1**, is bound permanently to the vehicle wheels **14** and **18** via the differential. Here, the axial flux machines **32***a* and **32***b*, for example, are formed as boost motors, which can each be switched on or off when driving forwards via the respective transfer element **42**, **44**, which is formed, in particular, as a centrifugal force freewheel, since no abutment of the clamping bodies acts when the boost motors **32***a, b* are inactive and thus no additional slipping losses and attrition on the freewheel. Wheel-individual drive torsional moment of the two boost motors **32***a* and **32***b* quickly make contact. Wheel-individual braking torsional moment cannot be directly generated when using freewheels as the transfer elements **42**, **44** with the two boost motors **32***a, b*), however this is possible indirectly. The central axial flux machine **32***c* provides, for example, the target braking moment. The boost motor **32***a* or *b* of the wheel side, which still requires drive moment, adds the required additional torsional moment, in order to obtain the required torsional moment at the wheel. The boost motors **32***a, b* have low switching times, since no active switching element is actuated and no synchronization of a switching element is required. Torsional moment, performance and highest speed requirements can thus advantageously also be depicted with one gear with this design of the embodiment of the invention. An ability to switch off the boost motors to the benefit of the highest speed is possible.
LIST OF REFERENCE CHARACTERS
(9) **10** Drive device **12** Output shaft **14** Vehicle wheel **16** Output shaft **18** Vehicle wheel **20** Differential transmission **22** Differential cage **24** Housing **26** Compensation wheel **28** Output wheel **30** Output wheel **32***a-c* Axial flux machine **34***a-c* Stator **36** Rotor **38** Rotor **40** Rotor **42** Transfer element **44** Transfer element **46** Bearing **48** Bearing **50** Planetary gearset **52** Planetary gearset **54**

Sun gear **56** Planetary carrier **58** Gear ring **60** Planetary wheel **62** Differential shaft **64** Differential shaft **66** Bearing **68** Bearing

Claims

1. A drive device (**10**) for a car, comprising: a first output shaft (**12**) via which a first vehicle wheel (**14**) of a vehicle axis of the car is drivable; a second output shaft (**16**) disposed coaxially to the first output shaft (**12**) via which a second vehicle wheel (**18**) of the vehicle axis of the car is drivable; a differential transmission (**20**) via which the first output shaft (**12**) and the second output shaft (**16**) are drivable; three axial flux machines (**32***a*-*c*) disposed coaxially to one another and coaxially to the first output shaft (**12**) and the second output shaft (**16**), wherein the three axial flux machines (**32***a*-*c*) are comprised by: a first axial flux machine (**32***a*) via which the first output shaft (**12**) is drivable by bypassing the differential transmission (**20**); a second axial flux machine (**32***b*) via which the second output shaft (**16**) is drivable by bypassing the differential transmission (**20**); and a third axial flux machine (**32***c*) disposed between the first axial flux machine (**32***a*) and the second axial flux machine (**32***b*) in an axial direction of the three axial flux machines (**32***a*-*c*) via which the first output shaft (**12**) and the second output shaft (**16**) are drivable via the differential transmission (**20**); a first switching element (**42**) formed as a first freewheel via which the first output shaft (**12**) is drivable by the first axial flux machine (**32***a*) by bypassing the differential transmission (**20**); and a second switching element (**44**) formed as a second freewheel (**44**) via which the second output shaft (**16**) is drivable by the second axial flux machine (**32***b*) by bypassing the differential transmission (**20**).
2. A drive device (**10**) for a car, comprising: a first output shaft (**12**) via which a first vehicle wheel (**14**) of a vehicle axis of the car is drivable; a second output shaft (**16**) disposed coaxially to the first output shaft (**12**) via which a second vehicle wheel (**18**) of the vehicle axis of the car is drivable; a differential transmission (**20**) via which the first output shaft (**12**) and the second output shaft (**16**) are drivable; three axial flux machines (**32***a*-*c*) disposed coaxially to one another and coaxially to the first output shaft (**12**) and the second output shaft (**16**), wherein the three axial flux machines (**32***a*-*c*) are comprised by: a first axial flux machine (**32***a*) via which the first output shaft (**12**) is drivable by bypassing the differential transmission (**20**); a second axial flux machine (**32***b*) via which the second output shaft (**16**) is drivable by bypassing the differential transmission (**20**); and a third axial flux machine (**32***c*) disposed between the first axial flux machine (**32***a*) and the second axial flux machine (**32***b*) in an axial direction of the three axial flux machines (**32***a*-*c*) via which the first output shaft (**12**) and the second output shaft (**16**) are drivable via the differential transmission (**20**); a first switching element (**42**) formed as a first frictionally engaged or positive-locking switching element via which the first output shaft (**12**) is drivable by the first axial flux machine (**32***a*) by bypassing the differential transmission (**20**); and a second switching element (**44**) formed as a second frictionally engaged or positive-locking switching element via which the second output shaft (**16**) is drivable by the second axial flux machine (**32***b*) by bypassing the differential transmission (**20**).
3. A drive device (**10**) for a car, comprising: a first output shaft (**12**) via which a first vehicle wheel (**14**) of a vehicle axis of the car is drivable; a second output shaft (**16**) disposed coaxially to the first output shaft (**12**) via which a second vehicle wheel (**18**) of the vehicle axis of the car is drivable; a differential transmission (**20**) via which the first output shaft (**12**) and the second output shaft (**16**) are drivable; three axial flux machines (**32***a*-*c*) disposed coaxially to one another and coaxially to the first output shaft (**12**) and the second output shaft (**16**), wherein the three axial flux machines (**32***a*-*c*) are comprised by: a first axial flux machine (**32***a*) via which the first output shaft (**12**) is drivable by bypassing the differential transmission (**20**); a second axial flux machine (**32***b*) via which the second output shaft (**16**) is drivable by bypassing the differential transmission (**20**); and a third axial flux machine (**32***c*) disposed between the first axial flux machine (**32***a*) and the second

axial flux machine (**32***b*) in an axial direction of the three axial flux machines (**32***a*-*c*) via which the first output shaft (**12**) and the second output shaft (**16**) are drivable via the differential transmission (**20**); a first switching element (**42**) formed as a first freewheel via which the first output shaft (**12**) is drivable by the third axial flux machine (**32***c*) by including the differential transmission (**20**); and a second switching element (**44**) formed as a second freewheel via which the second output shaft (**16**) is drivable by the third axial flux machine (**32***c*) by including the differential transmission (**20**).

4. A drive device (**10**) for a car, comprising: a first output shaft (**12**) via which a first vehicle wheel (**14**) of a vehicle axis of the car is drivable; a second output shaft (**16**) disposed coaxially to the first output shaft (**12**) via which a second vehicle wheel (**18**) of the vehicle axis of the car is drivable; a differential transmission (**20**) via which the first output shaft (**12**) and the second output shaft (**16**) are drivable; three axial flux machines (**32***a*-*c*) disposed coaxially to one another and coaxially to the first output shaft (**12**) and the second output shaft (**16**), wherein the three axial flux machines (**32***a*-*c*) are comprised by: a first axial flux machine (**32***a*) via which the first output shaft (**12**) is drivable by bypassing the differential transmission (**20**); a second axial flux machine (**32***b*) via which the second output shaft (**16**) is drivable by bypassing the differential transmission (**20**); and a third axial flux machine (**32***c*) disposed between the first axial flux machine (**32***a*) and the second axial flux machine (**32***b*) in an axial direction of the three axial flux machines (**32***a*-*c*) via which the first output shaft (**12**) and the second output shaft (**16**) are drivable via the differential transmission (**20**); a first switching element (**42**) formed as a first frictionally engaged or positive-locking switching element via which the first output shaft (**12**) is drivable by the third axial flux machine (**32***c*) by including the differential transmission (**20**); and a second switching element (**44**) formed as a second frictionally engaged or positive-locking switching element via which the second output shaft (**16**) is drivable by the third axial flux machine (**32***c*) by including the differential transmission (**20**).
